Unified Neuro-Adaptive Sliding Mode Control of Hexarotors with Unknown Disturbances

Mohsen Zahmatkesh, Honghao Pan, Maryam Zahmatkesh, Fatemeh Rekabi-Bana, Junyan Hu, and Farshad Arvin

2025 European Conference on Mobile Robots (ECMR) | 979-8-3315-2705-1/25/$31.00 ©2025 IEEE | DOI: 10.1109/ECMR65884.2025.11163349

***Abstract*— Intelligent control has been extensively applied to aerial robots, which is important for autonomous flight applications. However, it is susceptible to noise and uncertainties when integrating different sensors and nonlinear aerial robot dynamics. This paper proposes a robust, neural network-based adaptive hybrid control framework for a unified position and orientation tracking in Hexarotors. The proposed framework is considered for receiving uncertain data from altitude and orientation sensors. It further tackles estimating these time-varying noises, uncertainties, and unmodeled dynamics by synergistically integrating a second-order sliding mode control (SMC) with radial basis function neural networks (RBFNNs). The RBFNN-augmented SMC dynamically adapts control commands, with stability proven via the Lyapunov method. These advancements ensure a simultaneous and accurate position and orientation tracking under varying conditions. The framework is validated through extensive software and hardware in the loop (SIL/HIL) simulations, showcasing significant improvements in tracking accuracy, disturbance rejection, and adaptability compared to traditional methods.**

I. Introduction

Recently, the industrial adoption and configuration diversity of autonomous mobile robots have significantly expanded, primarily due to their inherently safer and smoother characteristics [1]. For the applications requiring high-precision maneuvers, such as precise manipulation [2], attitude control [3]–[6], and position control [7], while simultaneously aiming to reduce reliance on expensive sensors, the role of robust control strategies has become increasingly critical.

In recent years, many robust control methods have been developed for aerial vehicles. For instance, an adaptive second-order sliding mode fault-tolerant controller was developed in [8] to ensure stable flight and smooth landing under rotor failure. It introduces a reaching law to reduce chattering while maintaining fast and accurate tracking. While in [9], a cascade sliding mode control (SMC) was utilised to handle position control. In another work, [10], an integral back-stepping controller for the position control loop and a nonlinear disturbance observer-based SMC was proposed for fault-tolerant purposes. Although the mentioned works focused on SMC for achieving robust and adaptive control, the substantial advantages of integrating online neural networks were not fully explored.

Despite these advancements, several studies have incorporated neural networks (NN) into robust control frameworks. For example, in [11], a position and attitude tracking sliding mode controller (SMC) was developed for a quadrotor under parametric uncertainties and external disturbances, where the SMC parameters were adaptively tuned using an NN. Regarding outer loop control, a novel adaptive NN-based outer controller was developed in [12] for quadrotors that ensures robust trajectory tracking despite having no access to the inner loop dynamics. It guarantees error convergence and outperforms other adaptive NN-based schemes through experimental validation. Another work considered both inner and outer loop control, where an adaptive fault-tolerant control strategy is proposed in [13], combining fast terminal sliding mode control (FTSMC) with an NN that approximates system uncertainties in the quadrotor's attitude dynamics. In another work, [14] proposed an NN–based backstepping controller to estimate model uncertainties, unmodeled dynamics, external disturbances, and actuator faults. A composite learning strategy was further employed to optimize the update rules of both the NN and disturbance observer, leading to enhanced control performance. Robust controllers have also been applied to specific missions such as attitude tracking [15], where Q-learning was combined with fuzzy logic to enable continuous, fault-tolerant control in the presence of uncertainties, atmospheric noises, and actuator faults. Even with these prior achievements, there remains a strong motivation to develop a unified NN-augmented SMC framework for hexarotors to handle simultaneously both inner and outer loops.

In this work, a unified second-order sliding mode controller (SMC) is developed to simultaneously regulate both the position and attitude of a hexarotor in the presence of unknown disturbances. A Radial Basis Function (RBF) neural network is employed to estimate four distinct disturbances acting on the roll, pitch, yaw, and altitude dynamics. Accordingly, a nonlinear dynamic model of the hexarotor is derived, establishing a direct relationship between noise-affected Euler angles and position coordinates. Furthermore, a combinatorial error formulation encompassing both translational and rotational motions is introduced, and the estimated disturbances are incorporated into a unified control law to ensure effective compensation and robust tracking performance. The proposed method is validated through Hardware-in-the-Loop

M. Zahmatkesh, H. Pan, F. Rekabi-Bana, J. Hu, and F. Arvin are with the Department of Computer Science, Durham University, Durham, UK. (e-mail: mohsen.zahmatkesh@durham.ac.uk)

M. Zahmatkesh is with the Department of Mathematics, The University of Alabama, Tuscaloosa, USA

Authorized licensed use limited to: University of Durham. Downloaded on January 12,2026 at 00:32:55 UTC from IEEE Xplore. Restrictions apply.

simulation.

II. Nonlinear Dynamic Modeling of a Hexarotor

This section presents the nonlinear dynamic equations for a hexarotor considering rigid-body mechanics in three-dimensional space (Fig. 1). Let $\mathbf{P} = [X, Y, Z]^T \in \mathbb{R}^3$ denote the position in the earth-fixed frame, and $\boldsymbol{\alpha} = [\Phi, \Theta, \Psi]^T \in \mathbb{T}^3$ denote its Euler angles. The body-attached coordinate system $\mathcal{F}_B$ connects to the global frame $\mathcal{F}_G$ through rotational transformations:

$$\dot{\mathbf{P}} = \mathbf{C}_B^G \mathbf{V}_B, \quad \dot{\boldsymbol{\alpha}} = \mathbf{L}\boldsymbol{\Omega}_B, \tag{1}$$

where $\mathbf{V}_B = [V_x, V_y, V_z]^T$ and $\boldsymbol{\Omega}_B = [\omega_x, \omega_y, \omega_z]^T$ are translational and rotational velocities in $\mathcal{F}_B$. The orientation matrix $\mathbf{C}_B^G \in \mathrm{SO}(3)$ follows the ZYX convention [16]:

$$\mathbf{C}_B^G = \mathbf{C}_\Psi \mathbf{C}_\Theta \mathbf{C}_\Phi, \tag{2}$$

where $\mathbf{C}_\Psi$, $\mathbf{C}_\Theta$, and $\mathbf{C}_\Phi$ denote principal rotations about the Z, Y, and X axes, respectively. Angular rate conversion employs the kinematic coupling matrix:

$$\mathbf{L} = \begin{bmatrix} 1 & \sin\Phi\tan\Theta & \cos\Phi\tan\Theta \\ 0 & \cos\Phi & -\sin\Phi \\ 0 & \sin\Phi\sec\Theta & \cos\Phi\sec\Theta \end{bmatrix}. \tag{3}$$

The translational dynamics in $\mathcal{F}_B$ account for inertial effects and gravitational forces:

$$m\dot{\mathbf{V}}_B + m[\boldsymbol{\Omega}_B\times]\mathbf{V}_B = \mathbf{F}_B + m\mathbf{G}_B, \tag{4}$$

where

$$[\boldsymbol{\Omega}_B\times] \equiv \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}, \quad \mathbf{G}_B = g\begin{bmatrix} -\sin\Theta \\ \cos\Theta\sin\Phi \\ \cos\Theta\cos\Phi \end{bmatrix}, \tag{5}$$

are the skew-symmetric matrix of rotational velocity and gravitational acceleration. Hence, rotational motion follows:

$$\mathbf{J}_B\dot{\boldsymbol{\Omega}}_B + [\boldsymbol{\Omega}_B\times]\mathbf{J}_B\boldsymbol{\Omega}_B = \mathbf{M}_B, \tag{6}$$

where $\mathbf{J}_B = \mathrm{diag}(J_{xx}, J_{yy}, J_{zz})$ is the diagonal inertia tensor.

Aerodynamic forces and moments originate from rotor thrusts. The collective thrust F_T and moment components $\mathbf{M}_B = [M_\Phi, M_\Theta, M_\Psi]^T$ relate to rotor angular velocities $\{\varpi_i\}_{i=1}^6$ via:

$$\begin{bmatrix} F_T \\ M_\Phi \\ M_\Theta \\ M_\Psi \end{bmatrix} = \begin{bmatrix} \kappa & \kappa & \kappa & \kappa & \kappa & \kappa \\ -\frac{\kappa L}{2} & -\kappa L & -\frac{\kappa L}{2} & \frac{\kappa L}{2} & \kappa L & \frac{\kappa L}{2} \\ -\frac{\kappa L\sqrt{3}}{2} & 0 & \frac{\kappa L\sqrt{3}}{2} & \frac{\kappa L\sqrt{3}}{2} & 0 & -\frac{\kappa L\sqrt{3}}{2} \\ -\zeta & \zeta & -\zeta & \zeta & -\zeta & \zeta \end{bmatrix} \begin{bmatrix} \varpi_1^2 \\ \varpi_2^2 \\ \varpi_3^2 \\ \varpi_4^2 \\ \varpi_5^2 \\ \varpi_6^2 \end{bmatrix}, \tag{7}$$

where κ is the rotor thrust constant, L the moment arm length, and ζ the yaw drag coefficient.

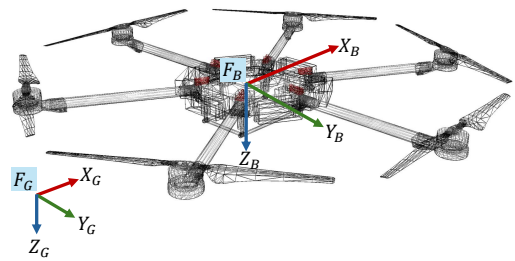

Fig. 1: Spatial reference frames and their geometric relationships

III. Robust Neural Adaptive Sliding Mode Control

The proposed controller ensures reliable tracking of position and attitude references $\boldsymbol{\chi}_d \triangleq [\boldsymbol{P}_d^\top, \boldsymbol{\alpha}_d^\top]^\top$ despite disturbances according to the block diagram 2. Define the state variables $\mathbf{x}_1 \triangleq \boldsymbol{\chi}$ and $\mathbf{x}_2 \triangleq \dot{\boldsymbol{\chi}}$ with dynamics:

$$\begin{cases} \dot{\mathbf{x}}_1 = \mathbf{B}_1\mathbf{x}_2 \\ \dot{\mathbf{x}}_2 = \mathbf{F}_2(\mathbf{x}_2) + \mathbf{B}_2(\mathbf{x}_2)\mathbf{u} + \boldsymbol{\Delta} \end{cases}, \tag{8}$$

where $\mathbf{B}_1 = \mathbf{I}_6$, $\boldsymbol{\Delta} \in \mathbb{R}^4$ represents lumped uncertainties, and under small-angle assumptions ($\|\phi\|, \|\theta\| < \pi/6$):

$$\mathbf{F}_2 = \begin{bmatrix} 0 \\ 0 \\ g \\ \frac{I_{yy}-I_{zz}}{I_{xx}}\dot{\theta}\dot{\psi} \\ \frac{I_{zz}-I_{xx}}{I_{yy}}\dot{\phi}\dot{\psi} \\ \frac{I_{xx}-I_{yy}}{I_{zz}}\dot{\phi}\dot{\theta} \end{bmatrix}, \ \mathbf{B}_2 = \begin{bmatrix} -\frac{s_\phi s_\psi + c_\phi c_\psi s_\theta}{m} & 0 & 0 & 0 \\ \frac{c_\phi s_\psi s_\theta - c_\psi s_\phi}{m} & 0 & 0 & 0 \\ -\frac{c_\phi c_\theta}{m} & 0 & 0 & 0 \\ 0 & \frac{l}{I_{xx}} & 0 & 0 \\ 0 & 0 & \frac{l}{I_{yy}} & 0 \\ 0 & 0 & 0 & \frac{l}{I_{zz}} \end{bmatrix}. \tag{9}$$

Lemma III.1. *For any continuous uncertainty $\boldsymbol{\Delta} : \mathbb{R}^6 \to \mathbb{R}^6$ on compact set $\Omega \subset \mathbb{R}^6$, there exists ideal weights $\mathbf{W}^* \in \mathbb{R}^{m\times 6}$ and Gaussian basis functions:*

$$\boldsymbol{\mu}(\mathbf{x}) = [e^{-\|\mathbf{x}-\mathbf{c}_1\|^2/2b_1^2}, \ldots, e^{-\|\mathbf{x}-\mathbf{c}_m\|^2/2b_m^2}]^\top, \tag{10}$$

such that $\boldsymbol{\Delta} = \mathbf{W}^{*\top}\boldsymbol{\mu}(\mathbf{x}) + \boldsymbol{\epsilon}$ with bounded approximation error $\|\boldsymbol{\epsilon}\| \leqslant \bar{\epsilon}$.

Assumption 1. *The uncertainty gradient satisfies $\|\nabla_{\mathbf{x}}\boldsymbol{\Delta}\| \leqslant L_\Delta$ over Ω.*

Proof: Following universal approximation theory [17], the optimal weights minimize $\|\mathbf{W}^\top\boldsymbol{\mu}(\mathbf{x}) - \boldsymbol{\Delta}(\mathbf{x})\|_{\mathcal{L}_\infty}$ on Ω.

Define tracking errors $\mathbf{e} \triangleq \mathbf{x}_1 - \mathbf{x}_{1_r}$ and $\dot{\mathbf{e}} \triangleq \mathbf{x}_2 - \mathbf{x}_{2_r}$ and construct the sliding manifold [11]:

$$\mathbf{s} \triangleq \dot{\mathbf{e}} + \boldsymbol{\Lambda}\mathbf{e}, \quad \boldsymbol{\Lambda} > 0, \tag{11}$$

With time derivative:

$$\dot{\mathbf{s}} = \ddot{\mathbf{e}} + \boldsymbol{\Lambda}\dot{\mathbf{e}}. \tag{12}$$

Assumption 2. *The combined uncertainty $\boldsymbol{\Delta}$ is differentiable with induced operator norm $\|\mathbf{D}\boldsymbol{\Delta}\| \leqslant L_\Delta$ over $\mathcal{X}$.*

Substituting system dynamics (8), yields:

$$\dot{\mathbf{s}} = \mathbf{F}_2 + \mathbf{B}_2\mathbf{u} + \boldsymbol{\Delta} - \ddot{\mathbf{x}}_{2_r} + \boldsymbol{\Lambda}\dot{\mathbf{e}}. \tag{13}$$

Authorized licensed use limited to: University of Durham. Downloaded on January 12,2026 at 00:32:55 UTC from IEEE Xplore. Restrictions apply.

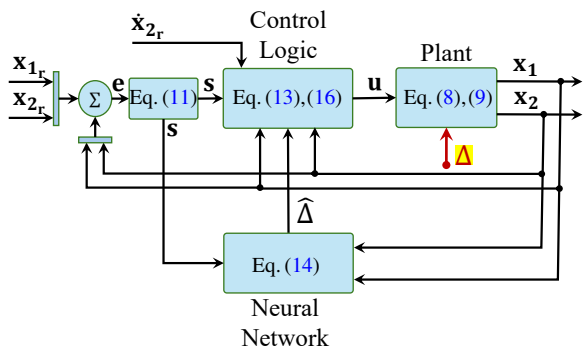

Fig. 2: Control diagram with references to equations.

Using the RBF approximation $\mathbf{\Delta} = \mathbf{W}^{*\top}\boldsymbol{\mu}(\mathbf{x}) + \boldsymbol{\epsilon}$ and its estimate:

$$\hat{\mathbf{\Delta}} = \hat{\mathbf{W}}^\top \boldsymbol{\mu}(\mathbf{x}), \tag{14}$$

the residual error satisfies:

$$\|\tilde{\mathbf{\Delta}}\| \leqslant \|\tilde{\mathbf{W}}\|_F \|\boldsymbol{\mu}\| + \bar{\epsilon}. \tag{15}$$

The control law could be derived by defining a reaching condition as $\dot{\mathbf{s}} = -\mathbf{K}\tanh\left(\frac{\mathbf{s}}{\Phi}\right)$ and using (13):

$$\mathbf{u} = \mathbf{B}_2^{-1}\left(-\mathbf{F}_2 - \hat{\mathbf{W}}^\top\boldsymbol{\mu} - \mathbf{\Lambda}\dot{\mathbf{e}} - \mathbf{K}\tanh\left(\frac{\mathbf{s}}{\Phi}\right) + \ddot{\mathbf{x}}_{2_r}\right). \tag{16}$$

Theorem III.2. *Under adaptation law* $\dot{\hat{\mathbf{W}}} = -\mathbf{\Gamma}\boldsymbol{\mu}(\mathbf{x})\mathbf{s}^\top$, *the sliding variable* $\mathbf{s}(t)$ *and the weight estimation error* $\tilde{\mathbf{W}}$ *are Uniformly Ultimately Bounded (UUB).*

Proof:

Using the Lyapunov function:

$$V = \frac{1}{2}\mathbf{s}^\top\mathbf{s} + \frac{1}{2}\operatorname{tr}\left(\tilde{\mathbf{W}}^\top\mathbf{\Gamma}^{-1}\tilde{\mathbf{W}}\right). \tag{17}$$

By differentiating along trajectories:

$$\dot{V} = \mathbf{s}^\top\dot{\mathbf{s}} + \operatorname{tr}(\tilde{\mathbf{W}}^\top\mathbf{\Gamma}^{-1}\dot{\hat{\mathbf{W}}}), \tag{18}$$

and by substituting (13):

$$\dot{V} = \mathbf{s}^\top\left(\mathbf{F}_2 + \mathbf{B}_2\mathbf{u} + \mathbf{\Delta} - \ddot{\mathbf{x}}_{2_r} + \mathbf{\Lambda}\dot{\mathbf{e}}\right) + \operatorname{tr}(\tilde{\mathbf{W}}^\top\mathbf{\Gamma}^{-1}\dot{\hat{\mathbf{W}}}), \tag{19}$$

where it can be expanded using (16):

$$\begin{aligned}\dot{V} = \mathbf{s}^\top\Big(&\mathbf{F}_2 + \mathbf{B}_2\mathbf{B}_2^{-1}\big(-\mathbf{F}_2 - \hat{\mathbf{\Delta}}(\mathbf{x}_2) - \mathbf{\Lambda}\dot{\mathbf{e}} - \mathbf{K}\tanh\left(\frac{\mathbf{s}}{\Phi}\right)\\ &+ \ddot{\mathbf{x}}_{2_r}\big) + \mathbf{\Delta}(\mathbf{x}_2) - \ddot{\mathbf{x}}_{2_r} + \mathbf{\Lambda}\dot{\mathbf{e}}\Big) + \operatorname{tr}(\tilde{\mathbf{W}}^\top\mathbf{\Gamma}^{-1}\dot{\hat{\mathbf{W}}}),\end{aligned} \tag{20}$$

which reduces to the compact form:

$$\dot{V} = \mathbf{s}^\top\left(\mathbf{\Delta} - \hat{\mathbf{\Delta}} - \mathbf{K}\tanh\mathbf{s}\right) + \operatorname{tr}(\tilde{\mathbf{W}}^\top\mathbf{\Gamma}^{-1}\dot{\hat{\mathbf{W}}}). \tag{21}$$

Now substitute the adaptation law:

$$\dot{\hat{\mathbf{W}}} = -\mathbf{\Gamma}\boldsymbol{\mu}\mathbf{s}^\top \quad \Rightarrow \quad \operatorname{tr}(\tilde{\mathbf{W}}^\top\mathbf{\Gamma}^{-1}\dot{\hat{\mathbf{W}}}) = -\mathbf{s}^\top\tilde{\mathbf{W}}^\top\boldsymbol{\mu}$$

Applying the weight error dynamics $\tilde{\mathbf{\Delta}} = \tilde{\mathbf{W}}^\top\boldsymbol{\mu} + \boldsymbol{\epsilon}$:

$$\begin{aligned}\dot{V} &= -\mathbf{s}^\top\mathbf{K}\tanh\left(\frac{\mathbf{s}}{\Phi}\right) + \mathbf{s}^\top\boldsymbol{\epsilon}\\ &\leqslant -\lambda_{\min}(\mathbf{K})\mathbf{s}^\top\tanh\left(\frac{\mathbf{s}}{\Phi}\right) + \bar{\epsilon}|\mathbf{s}|\\ &\leqslant -\lambda_{\min}(\mathbf{K})\left(|\mathbf{s}| - n\Phi\ln 2\right) + \bar{\epsilon}|\mathbf{s}|,\end{aligned}$$

where the final inequality follows from the component-wise bound $s_i\tanh(s_i/\Phi) \geqslant |s_i| - \Phi\ln 2$. This guarantees $\dot{V} < 0$ when $|\mathbf{s}| > \frac{\lambda_{\min}(\mathbf{K})n\Phi\ln 2}{\lambda_{\min}(\mathbf{K})-\bar{\epsilon}}$, proving uniform ultimate boundedness.

A. Neural Adaptive Altitude Control

Following the established framework, define altitude tracking errors as $\varepsilon_z = z - z_r$ and $\dot{\varepsilon}_z = \dot{z} - \dot{z}_r$. The vertical sliding manifold and its derivative become:

$$s_z = \dot{\varepsilon}_z + \gamma_z\varepsilon_z, \quad \dot{s}_z = \ddot{\varepsilon}_z + \gamma_z\dot{\varepsilon}_z, \tag{22}$$

where $\gamma_z > 0$ governs convergence rate. Utilizing the dynamics from (13) and substituting (9) in it, the manifold derivative expands to:

$$\dot{s}_z = g - \frac{\cos\phi\cos\theta}{m}u_1 + \Delta_z - \ddot{z}_r + \gamma_z\dot{\varepsilon}_z. \tag{23}$$

Synthesizing the control input via the prescribed reaching law (16) yields:

$$u_1 = \frac{m}{\cos\phi\cos\theta}\left(g + \hat{\Delta}_z + \gamma_z\dot{\varepsilon}_z + \kappa_z\tanh\left(\frac{s_z}{\Phi_z}\right) - \ddot{z}_r\right). \tag{24}$$

B. Neural Adaptive Directional Control

Define yaw orientation errors as $\varepsilon_\psi = \psi - \psi_r$ and $\dot{\varepsilon}_\psi = \dot{\psi} - \dot{\psi}_r$. The directional sliding manifold and its derivative are constructed as:

$$s_\psi = \dot{\varepsilon}_\psi + \gamma_\psi\varepsilon_\psi, \quad \dot{s}_\psi = \ddot{\varepsilon}_\psi + \gamma_\psi\dot{\varepsilon}_\psi, \tag{25}$$

with $\gamma_\psi > 0$ determining angular convergence. Substituting rotational dynamics from (13), and (9) produces:

$$\dot{s}_\psi = \frac{I_{xx} - I_{yy}}{I_{zz}}\dot{\phi}\dot{\theta} + \frac{l}{I_{zz}}u_4 + \Delta_\psi - \ddot{\psi}_r + \gamma_\psi\dot{\varepsilon}_\psi. \tag{26}$$

Leveraging the adaptation mechanism from (16), the yaw control command becomes:

$$\begin{aligned}u_4 = \frac{I_{zz}}{l}\Bigg(&-\frac{I_{xx} - I_{yy}}{I_{zz}}\dot{\phi}\dot{\theta} - \kappa_\psi\tanh\left(\frac{s_\psi}{\Phi_\psi}\right) - \hat{\Delta}_\psi\\ &+ \ddot{\psi}_r - \gamma_\psi\dot{\varepsilon}_\psi\Bigg).\end{aligned} \tag{27}$$

C. Neural Adaptive Longitudinal and Pitch Control

In this section, a combinatorial error is defined for pitch orientation θ and longitudinal position x tracking in the presence of unknown disturbances. Define the composite sliding manifold:

$$s_{\theta x} = \gamma_\theta(\theta - \theta_r) + \gamma_{\dot{\theta}}(\dot{\theta} - \dot{\theta}_r) + \gamma_x(x - x_r) + \gamma_{\dot{x}}(\dot{x} - \dot{x}_r), \tag{28}$$

where $\gamma_\theta, \gamma_{\dot{\theta}}, \gamma_x, \gamma_{\dot{x}} > 0$ are design parameters. Hence, the manifold dynamics becomes:

$$\dot{s}_{\theta x} = \gamma_\theta(\dot{\theta} - \dot{\theta}_r) + \gamma_{\dot{\theta}}(\ddot{\theta} - \ddot{\theta}_r) + \gamma_x(\dot{x} - \dot{x}_r) + \gamma_{\dot{x}}(\ddot{x} - \ddot{x}_r). \tag{29}$$

Authorized licensed use limited to: University of Durham. Downloaded on January 12,2026 at 00:32:55 UTC from IEEE Xplore. Restrictions apply.

Enforcing sliding manifolds $s_{\theta x} = 0$, $\dot{s}_{\theta x} = 0$, and defining state variables $z_1 = \theta - \theta_r$, $z_2 = \dot{\theta} - \dot{\theta}_r$, and $z_3 = x - x_r$, the system admits the state-space representation:

$$\begin{cases} \dot{z}_1 = z_2 \\ \dot{z}_2 = \left(-\frac{\gamma_\theta}{\gamma_{\dot{\theta}}} + \frac{\gamma_x}{\gamma_{\dot{x}}}\right) z_2 + \frac{\gamma_x \gamma_\theta}{\gamma_{\dot{\theta}} \gamma_{\dot{x}}} z_1 + \frac{\gamma_x^2}{\gamma_{\dot{\theta}} \gamma_{\dot{x}}} z_3 \\ \quad + \frac{\gamma_{\dot{x}}}{\gamma_{\dot{\theta}} m}(s_\phi s_\psi + c_\phi c_\psi s_{(\theta_r + z_1)}) u_1 - \frac{\gamma_{\dot{x}}}{\gamma_{\dot{\theta}}} \Delta_\theta + \frac{\gamma_{\dot{x}}}{\gamma_{\dot{\theta}}} \ddot{x}_r \\ \dot{z}_3 = -\frac{\gamma_\theta}{\gamma_{\dot{x}}} z_1 - \frac{\gamma_{\dot{\theta}}}{\gamma_{\dot{x}}} z_2 - \frac{\gamma_x}{\gamma_{\dot{x}}} z_3. \end{cases} \tag{30}$$

Linearization about the equilibrium $(z_1, z_2, z_3) = (0, 0, 0)$ produces the Jacobian:

$$\mathbf{J} = \begin{bmatrix} 0 & 1 & 0 \\ \frac{\gamma_x \gamma_\theta}{\gamma_{\dot{\theta}} \gamma_{\dot{x}}} + \frac{\gamma_{\dot{x}}}{\gamma_{\dot{\theta}} m} c_\phi c_\psi c_{\theta_r} u_1 & -\frac{\gamma_\theta}{\gamma_{\dot{\theta}}} + \frac{\gamma_x}{\gamma_{\dot{x}}} & \frac{\gamma_x^2}{\gamma_{\dot{\theta}} \gamma_{\dot{x}}} \\ -\frac{\gamma_\theta}{\gamma_{\dot{x}}} & -\frac{\gamma_{\dot{\theta}}}{\gamma_{\dot{x}}} & -\frac{\gamma_x}{\gamma_{\dot{x}}} \end{bmatrix}. \tag{31}$$

The characteristic polynomial with coefficients $\omega_1, \omega_2, \omega_3$:

$$s^3 + \omega_3 s^2 + \omega_1 s + \omega_2 = 0, \tag{32}$$

satisfies asymptotic stability per Routh-Hurwitz when $\omega_i > 0$ and $\omega_2 < \omega_1 \omega_3$. Practical implementation uses the parameterization:

$$\omega_3 = \frac{\gamma_\theta}{\gamma_{\dot{\theta}}}, \quad \omega_1 = -\frac{\gamma_{\dot{x}}}{\gamma_{\dot{\theta}} m} c_\phi c_\psi c_{\theta_r} u_1, \quad \omega_2 = -\frac{\gamma_x}{\gamma_{\dot{\theta}} m} c_\phi c_\psi c_{\theta_r} u_1, \tag{33}$$

and the control input can be generated from (29) and (16) by using (8), and (9) as:

$$u_3 = \frac{I_{yy}}{l}\Bigg(-\frac{1}{\gamma_{\dot{\theta}}} \kappa_\theta \tanh\left(\frac{s_{\theta x}}{\Phi}\right) - \frac{\gamma_\theta}{\gamma_{\dot{\theta}}} z_2 - \frac{I_{zz} - I_{xx}}{I_{yy}} \dot{\phi}\dot{\psi} \\ - \Delta_\theta + \ddot{\theta}_r - \frac{\gamma_x}{\gamma_{\dot{\theta}}} \dot{z}_3 + \frac{\gamma_{\dot{x}}}{\gamma_{\dot{\theta}}} \left(\frac{s_\phi s_\psi + c_\phi c_\psi s_\theta}{m} u_1 + \ddot{x}_r \right) \Bigg). \tag{34}$$

D. Neural Adaptive Lateral and Roll Control

Similarly, a combinatorial error and neural adaptive sliding mode control law for lateral position y and roll angle ϕ is developed in this section:

$$s_{\phi y} = \gamma_\phi(\phi - \phi_r) + \gamma_{\dot{\phi}}(\dot{\phi} - \dot{\phi}_r) + \gamma_y(y - y_r) + \gamma_{\dot{y}}(\dot{y} - \dot{y}_r), \tag{35}$$

and the manifold dynamics become:

$$\dot{s}_{\phi y} = \gamma_\phi(\dot{\phi} - \dot{\phi}_r) + \gamma_{\dot{\phi}}(\ddot{\phi} - \ddot{\phi}_r) + \gamma_y(\dot{y} - \dot{y}_r) + \gamma_{\dot{y}}(\ddot{y} - \ddot{y}_r). \tag{36}$$

Enforcing sliding manifolds $s_{\phi y} = 0$, $\dot{s}_{\phi y} = 0$, where $z_4 = \phi - \phi_r$, $z_5 = \dot{\phi} - \dot{\phi}_r$, $z_6 = y - y_r$, produces:

$$\begin{cases} \dot{z}_4 = z_5 \\ \dot{z}_5 = -\frac{\gamma_\phi}{\gamma_{\dot{\phi}}} z_5 + \frac{\gamma_y \gamma_\phi}{\gamma_{\dot{\phi}} \gamma_{\dot{y}}} z_4 + \frac{\gamma_y}{\gamma_{\dot{y}}} z_5 + \frac{\gamma_y^2}{\gamma_{\dot{\phi}} \gamma_{\dot{y}}} z_6 \\ \quad - \frac{\gamma_{\dot{y}}}{\gamma_{\dot{\phi}}} \left(\frac{c_\phi s_\psi s_\theta - c_\psi s_{(\phi_r + z_4)}}{m} u_1 - \ddot{y}_r \right) \\ \dot{z}_6 = -\frac{1}{\gamma_{\dot{y}}} \left(\gamma_\phi z_4 + \gamma_{\dot{\phi}} z_5 + \gamma_y z_6 \right), \end{cases} \tag{37}$$

where linearizing about the equilibrium $(z_4, z_5, z_6) = (0, 0, 0)$:

$$\mathbf{J} = \begin{bmatrix} 0 & 1 & 0 \\ \frac{\gamma_\phi \gamma_y}{\gamma_{\dot{\phi}} \gamma_{\dot{y}}} + \frac{c_\psi \gamma_{\dot{y}} u_1 c_{\phi_r}}{\gamma_{\dot{\phi}} m} & \frac{\gamma_y}{\gamma_{\dot{y}}} - \frac{\gamma_\phi}{\gamma_{\dot{\phi}}} & \frac{\gamma_y^2}{\gamma_{\dot{\phi}} \gamma_{\dot{y}}} \\ -\frac{\gamma_\phi}{\gamma_{\dot{y}}} & -\frac{\gamma_{\dot{\phi}}}{\gamma_{\dot{y}}} & -\frac{\gamma_y}{\gamma_{\dot{y}}} \end{bmatrix}. \tag{38}$$

The linearized system's characteristic polynomial with stability parameters $\omega_4, \omega_5, \omega_6$:

$$s^3 + \omega_6 s^2 + \omega_4 s + \omega_5 = 0, \tag{39}$$

achieves asymptotic stability when $\omega_4, \omega_5, \omega_6 > 0$ and $\omega_5 < \omega_4 \omega_6$ through the parameter mapping. This is satisfied through the parameter mapping:

$$\omega_6 = \frac{\gamma_\phi}{\gamma_{\dot{\phi}}}, \quad \omega_4 = -\frac{c_\psi \gamma_{\dot{y}} u_1 c_{\phi_r}}{\gamma_{\dot{\phi}} m}, \quad \omega_5 = -\frac{c_\psi \gamma_y u_1 c_{\phi_r}}{\gamma_{\dot{\phi}} m}. \tag{40}$$

Substituting the lateral dynamics from (8), and (9), while counting on the reaching condition $\dot{s}_{\phi y} = -\kappa_\phi \tanh\left(\frac{s_{\phi y}}{\Phi_y}\right)$ in (16), yields the control command as follows:

$$u_2 = \frac{I_{xx}}{l}\Bigg(-\frac{1}{\gamma_{\dot{\phi}}} \kappa_\phi \tanh\left(\frac{s_{\phi y}}{\Phi_y}\right) - \frac{\gamma_\phi}{\gamma_{\dot{\phi}}} z_5 - \frac{I_{yy} - I_{zz}}{I_{xx}} \dot{\theta}\dot{\psi} \\ - \Delta_\phi + \ddot{\phi}_r - \frac{\gamma_y}{\gamma_{\dot{\phi}}} \dot{z}_6 - \frac{\gamma_{\dot{y}}}{\gamma_{\dot{\phi}}} \left(\frac{c_\phi s_\psi s_\theta - c_\psi s_\phi}{m} u_1 - \ddot{y}_r \right) \Bigg). \tag{41}$$

IV. Simulation Results

This section presents the simulation results of the neural adaptive SMC method. A 75 seconds square flight trajectory, including the takeoff and landing phases, is defined. The reference position commands for the quadrotor along the x, y, and z axes are given as follows:

$$x_r(t) = \begin{cases} 0 & 0 \leqslant t < 15 \\ \frac{t-15}{15} & 15 \leqslant t < 30 \\ 1 & 30 \leqslant t < 45 \\ 1 - \frac{t-45}{15} & 45 \leqslant t \leqslant 60 \\ 0 & t > 60 \end{cases} \qquad z_r(t) = \begin{cases} 2 & 0 \leqslant t < 60 \\ 0 & t \geqslant 60 \end{cases}$$

$$y_r(t) = \begin{cases} \frac{t}{15} & 0 \leqslant t < 15 \\ 1 & 15 \leqslant t < 30 \\ 1 - \frac{t-30}{15} & 30 \leqslant t < 45 \\ 0 & 45 \leqslant t \leqslant 60 \\ 0 & t > 60 \end{cases} \tag{42}$$

The disturbances implemented on the four states consist of a constant deviation term and a time-varying term. The total disturbance vector $\mathbf{\Delta}(t)$ is expressed:

$$\boldsymbol{d_1}(t) = \begin{bmatrix} \phi(t)\,\theta(t) \\ \theta(t)\,\psi(t) \\ \phi(t)\,\psi(t) \\ 0.2\,\dot{z}(t) \end{bmatrix}, \qquad \mathbf{d_2}(t) = \begin{bmatrix} 5\sin\left(\frac{\pi t}{2}\right) \\ 5\sin\left(\frac{\pi t}{3}\right) \\ 2\sin(\pi t) \\ 6\sin\left(2\pi t + \frac{\pi}{6}\right) \end{bmatrix}, \tag{43}$$

$$\mathbf{\Delta}(t) = \boldsymbol{d_1}(t) + \mathbf{d_2}(t). \tag{44}$$

Authorized licensed use limited to: University of Durham. Downloaded on January 12,2026 at 00:32:55 UTC from IEEE Xplore. Restrictions apply.

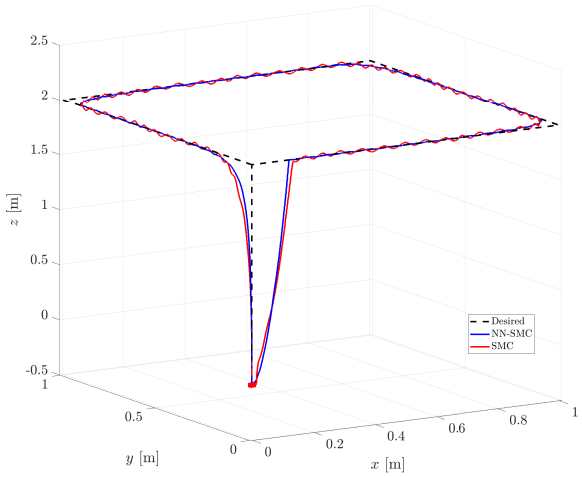

Fig. 3: 3D trajectory tracking using the proposed RBF-SMC method versus traditional SMC.

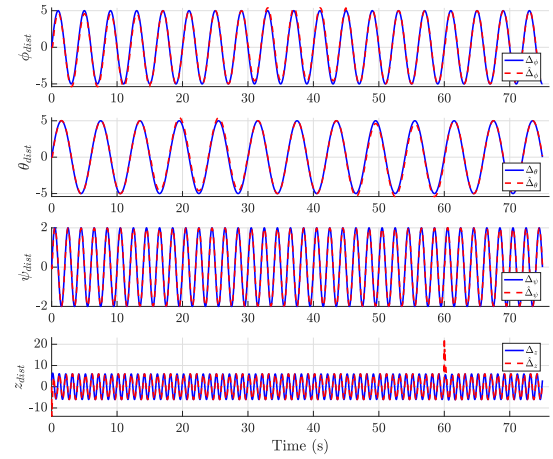

Fig. 4: Disturbance estimation performance of the RBF neural network.

Here, $\boldsymbol{d_1}(t)$ represents the state-dependent disturbance, while $\mathbf{d_2}(t)$ captures the time-varying sinusoidal noise. The complete simulation setup is available in GitHub. The control parameters are selected for the sliding mode controllers applied to the altitude, yaw, pitch, and roll dynamics. These parameters include the adaptation gain (Γ), sliding surface slopes (κ_z, κ_ψ, κ_θ, κ_ϕ), and control gains. All parameters are tuned to ensure system stability, accurate tracking performance, and robustness against model uncertainties and external disturbances.

According to Fig. 3, the performance of the proposed RBF-SMC (Radial Basis Function Sliding Mode Control) method surpasses that of the traditional SMC approach without the use of a neural network. Specifically, the RBF-SMC method achieves lower tracking error and reduced control effort compared to the conventional SMC. It is also evident that the implemented noises make a harsh landing during the last seconds.

Figure 4 demonstrates the estimation capabilities of the RBF neural network, configured with 40 hidden neurons, 14 first-layer neurons, and 4 output neurons estimating four states. The results confirm that all four disturbances are accurately estimated and effectively incorporated into the control loop, enhancing the robustness of the system.

The effectiveness of the RBF-SMC method extends beyond position and attitude tracking. As illustrated in Fig. 5, the linear velocity components in the inertial frame also benefit from the adaptive control strategy. The RBF-SMC approach demonstrates improved stability and responsiveness.

Furthermore, the superiority of the RBF-SMC method is especially evident in attitude control, as shown in Fig. 6. The variations in roll (ϕ) and pitch (θ) reflect directional changes during the square trajectory, yet the RBF-SMC maintains greater stability. The most pronounced improvement is observed in the yaw angle (ψ), where the traditional SMC exhibits poor performance, while the proposed method provides significantly smoother and more accurate tracking.

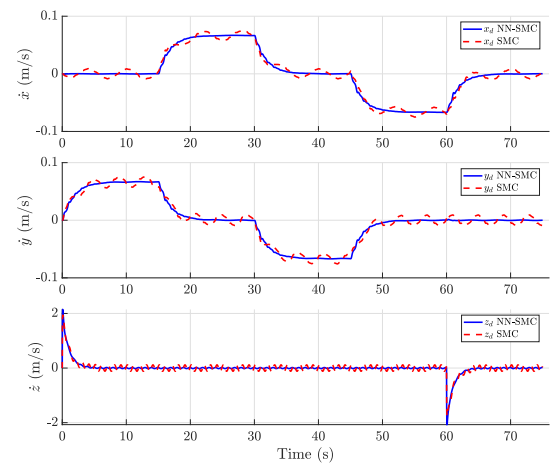

Fig. 5: Linear velocity profiles of the RBF-SMC method versus the traditional SMC.

V. Hardware in the Loop

To evaluate the practicality of the proposed approach, an experimental test was conducted using an ESP32-WROOM-32 board. As shown in Fig.7, the Hexarotor 6 DoF dynamics and simulation were implemented in Simulink 2024b on a PC with an Intel(R) Core(TM) i9-14900HX (24 cores, 32 threads, 2.2 GHz). At each timestep, system states were sent to the ESP32 via serial communication (115200 bps) using the Simulink Support Package for Arduino. The ESP32 then computed control commands in real time to validate the controller's performance. Figure 8 illustrates the effect of the hardware-in-the-Loop simulation on the control commands. The results confirm that the proposed controller successfully generates control inputs while maintaining system stability.

VI. Conclusion

This work presents a unified second-order sliding mode controller (SMC) for hexarotor position and attitude control, verified through Hardware-in-the-Loop simulation. A Radial Basis Function (RBF) neural network estimates unknown

Authorized licensed use limited to: University of Durham. Downloaded on January 12,2026 at 00:32:55 UTC from IEEE Xplore. Restrictions apply.

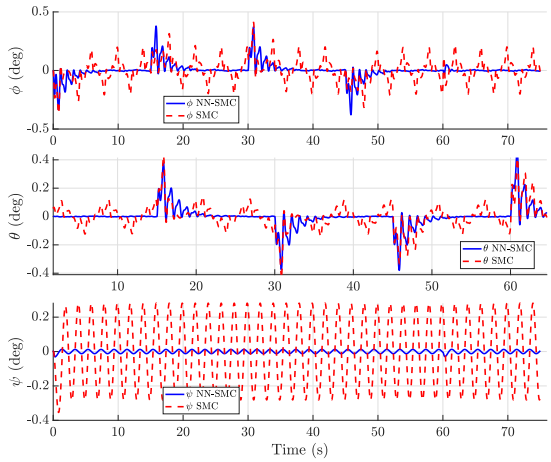

Fig. 6: Euler angle response comparison between the RBF-SMC and traditional SMC methods.

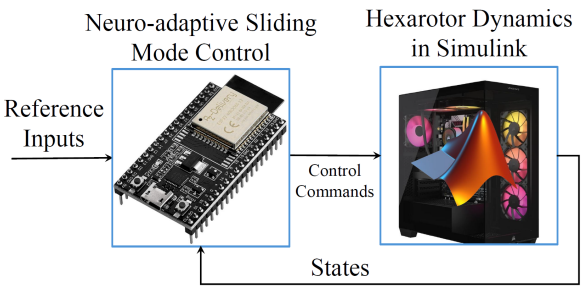

Fig. 7: Hardware-in-the-loop (HIL) schematic.

disturbances in the roll, pitch, yaw, and altitude channels. The controller successfully maintains stability and precision despite challenging, simultaneous disturbances on both linear and rotational dynamics, demonstrating its robustness and the effectiveness of the real-time RBF disturbance estimation.

References

[1] S. Rajappa, M. Ryll, H. H. Bülthoff, and A. Franchi, "Modeling, control and design optimization for a fully-actuated hexarotor aerial vehicle with tilted propellers," in *2015 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 4006–4013, 2015.

[2] J. Ulrich, M. Stefanec, F. Rekabi-Bana, *et al.*, "Autonomous tracking of honey bee behaviors over long-term periods with cooperating robots," *Science Robotics*, vol. 9, no. 95, p. eadn6848, 2024.

[3] J. Li and L. Wu, "A multiple connected recurrent neural network based super-twisting terminal sliding mode control for quad-rotor uav," *European Journal of Control*, vol. 83, p. 101220, 2025.

[4] B. Li, M. Chen, S. Qi, and K. Peng, "Finite-time fault-tolerant control of attitude control system of quadrotor uav based on neural network disturbance observer," *Neural Computing and Applications*, vol. 37, no. 7, pp. 5597–5606, 2025.

[5] H. Hassani, A. Mansouri, and A. Ahaitouf, "Design and real-time implementation of a robust backstepping non-singular integral terminal sliding mode attitude control for a quadrotor aircraft," *Journal of Vibration and Control*, vol. 31, no. 7-8, pp. 1278–1293, 2025.

[6] S. Ranjan and S. Majhi, "Fixed-time state observer-based robust adaptive neural fault-tolerant control for a quadrotor unmanned aerial vehicle," *International Journal of Adaptive Control and Signal Processing*, vol. 39, no. 1, pp. 132–151, 2025.

[7] A. M. Aghazamani and M. Khodabandeh, "Position tracking control of a quadrotor using a fuzzy adaptive feedback linearization controller," *Journal of Control, Automation and Electrical Systems*, vol. 36, no. 2, pp. 193–206, 2025.

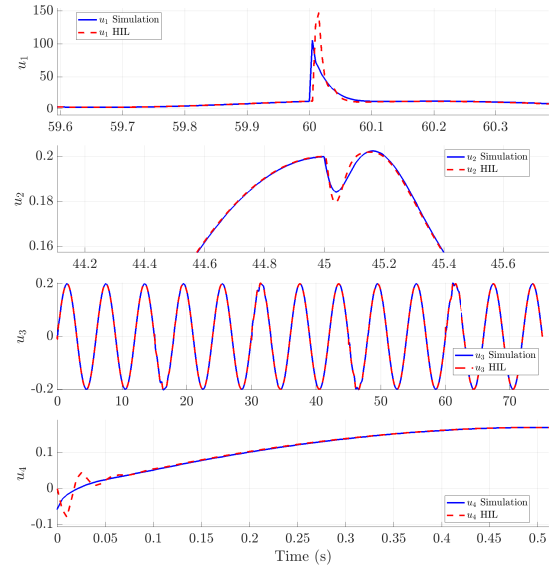

Fig. 8: Control inputs u_1 to u_4 generated by the NN-SMC controller in simulation compared with the HIL.

[8] S. Lian, Y. Zhu, W. Meng, K. Shao, and H. Li, "Adaptive fault-tolerant control for quadrotor based on the second-order fast nonsingular terminal sliding mode control," *IEEE Transactions on Industrial Electronics*, pp. 1–11, 2025.

[9] A. Sawiński, P. Chudzik, and K. Tatar, "Cascade sliding mode control for linear displacement positioning of a quadrotor," *Sensors (Basel, Switzerland)*, vol. 25, no. 3, p. 883, 2025.

[10] M. Abdullah, A. Zulfiqar, and J. H. Arman, "An integral backstepping and nonlinear disturbance observer-based sliding mode control for active fault-tolerant quadcopter uavs with simulation as well as hardware-in-the-loop experiments," *International Journal of Dynamics and Control*, vol. 13, no. 4, p. 159, 2025.

[11] H. Razmi and S. Afshinfar, "Neural network-based adaptive sliding mode control design for position and attitude control of a quadrotor uav," *Aerospace Science and Technology*, vol. 91, pp. 12–27, 2019.

[12] I. Lopez-Sanchez, J. Moyrón, and J. Moreno-Valenzuela, "Adaptive neural network-based trajectory tracking outer loop control for a quadrotor," *Aerospace Science and Technology*, vol. 129, p. 107847, 2022.

[13] B. Gao, Y.-J. Liu, and L. Liu, "Adaptive neural fault-tolerant control of a quadrotor uav via fast terminal sliding mode," *Aerospace Science and Technology*, vol. 129, p. 107818, 2022.

[14] M. Ezabadi, M. Zahmatkesh, S. A. Emami, and P. Castaldi, "Robust neuro-adaptive command-filtered back-stepping fault-tolerant control of satellite using composite learning," *Advances in Space Research*, vol. 75, no. 1, pp. 1231–1244, 2025.

[15] M. Zahmatkesh, S. A. Emami, A. Banazadeh, and P. Castaldi, "Robust attitude control of an agile aircraft using improved q-learning," in *Actuators*, vol. 11, p. 374, MDPI, 2022.

[16] M. Ilyas, Z. Feng, S. Ullah, and J. J. Liu, "Pso-nonsingular fast terminal sliding mode control for hexarotor in turbulent atmospheric boundary layer with disturbance observer," *International Journal of Robust and Nonlinear Control*, 2025.

[17] G. Cybenko, "Approximation by superpositions of a sigmoidal function," *Mathematics of control, signals and systems*, vol. 2, no. 4, pp. 303–314, 1989.

Authorized licensed use limited to: University of Durham. Downloaded on January 12,2026 at 00:32:55 UTC from IEEE Xplore. Restrictions apply.